

K803 — Diagnosis of why the capstone stalled: the team conflated RADIAL (bulk interior) with BOUNDARY ORBITS. Casey's fix: a quark is CONFINED = an interior bulk state ("rolling in the curvature," not emitted); a lepton is colorless = a boundary (Shilov) state. TWO different address structures, not one. This reconnects to the established bulk-vs-Shilov two-region structure the team drifted from — and explains why the boundary-orbit Gindikin norms didn't output 20 (quarks aren't at the boundary orbits). Redo the QUARK computation as a confined interior O-overlap. Same validation gates.

Keeper | 2026-07-21 Tuesday | Casey: "people don't know if they're doing radial or boundary orbits. What's a quark — it's confined, a state inside rolling in the curvature, not emitted." Corpus-grounded (bulk-vs-Shilov, Casey-elevated) + web-grounded (holographic QCD confinement = bulk modes).

The diagnosis (Casey) — the two structures were conflated

The capstone stalled on a confusion Lyra surfaced: two candidate address structures floating around — boundary orbits $\{0,1,2\}$ (Korányi–Wolf strata, Shilov) vs interior radii $\{0, \frac{1}{4}, \frac{2}{3}\}$ ($r^2 < 1$, bulk, Grace's $r^2 = k/(k+N_c)$). The team used them interchangeably. Casey's physics resolves it: - Quark = CONFINED ∇ an interior BULK state — normalizable, square-integrable, "rolling around in the curvature," CANNOT reach/be emitted at the boundary. → the interior radii $\{0, \frac{1}{4}, \frac{2}{3}\}$ picture (with color N_c). - Lepton = colorless ∇ reaches the Shilov BOUNDARY — free, emitted. → the boundary-orbit strata picture. - TWO distinct address structures. The team wrongly put quarks at the boundary orbits (like leptons). That is why the naive boundary-orbit Gindikin norms ($\{5,6,17.5,21,27\}$ Elie / $\{2.5,7,13.5,22\}$ Lyra) did NOT output 20 — quarks aren't there.

This reconnects to ESTABLISHED BST structure (not a new speculation)

CLAUDE.md, Casey-elevated: "Bulk-vs-Shilov two-region: Shilov boundary = light fundamental particles (leptons); Bulk interior = confined composite particles (quarks)." The address-pinning work DRIFTED from this durable result. Casey is reconnecting it. Confidence-raising: the fix is a return to a banked structure, not a fresh guess.

Web grounding — confinement = normalizable bulk modes; D_{IV}^5 boundedness IS the cutoff

Holographic QCD (light-front, Brodsky–de Téramond): confinement = discrete NORMALIZABLE bulk modes, obtained by truncating the space in the IR; hadron emission = boundary operators. D_{IV}^5 is a BOUNDED domain → automatically IR-truncated → its interior (bulk) modes ARE normalizable discrete-series = confined, with no cutoff by hand. The boundedness of D_{IV}^5 is the confinement mechanism. So: quark = interior discrete-series bulk mode; lepton/emission = Shilov boundary. Physics and geometry agree.

★ The investigation — redo the QUARK computation as a confined interior O-overlap

- Separate the two structures cleanly: leptons at the boundary orbits (Shilov strata — where the clean lepton ladder T190/T2003 lives); quarks at the interior radii $r^2 = k/(k+N_c)$ (bulk, colored, confined).
- The quark mass is the interior O-overlap $\langle f_n | O | f_n \rangle$ (Elie's point — NOT the bare norm ratio): a confined interior mode overlapped with the boundary-reaching condensate O, at interior radius $r^2 = k/(k+N_c)$, weighted by the bulk measure AND the color N_c structure. This is naturally "how close does the confined quark get to the boundary."

- ★ Gate (a), redone in the right frame: does the confined interior O-overlap output $m_s/m_d = 20$ tuning-free (derived ν , no ad-hoc shift)? The boundary-orbit version failed; the interior/confined version is the physically-correct computation — do it.
- Refined top (refines F603 saturation): the top is the nearest-boundary CONFINED quark — the largest interior overlap with O, but still $r^2 < 1$ (confined), NOT literally O at the boundary. “Saturation” = nearest-interior-to-the-wall, not reaching it. (A confined quark can’t BE the boundary condensate.) More physical.

Discipline — this is the THIRD reframe; the gates are UNCHANGED

Capstone frames so far: search → determined-computation → confined-interior. Each felt promising; two didn’t validate. The confinement reframe improves the STARTING FRAME (and it’s corpus+web grounded), but it is NOT a validation. Bank nothing until: - (a) the confined interior O-overlap OUTPUTS $m_s/m_d=20$ tuning-free, AND - (b) a SECOND independent quantity ($V_{cb} / m_c/m_u / \nu \Delta m^2$) is PREDICTED and checked. - Sub-check (Grace’s hinge, still live): is gen-3’s interior level $k=C_2=6$ FORCED (e.g. the last normalizable K-type before the Wallach threshold — a real candidate) or fit to land $r^2=2/3$? Load-bearing. - Pivot-exit stands: if the interior computation also needs ad-hoc tuning, or $k=6$ is a fit, bank the flavor-synthesis paper and pivot.

What this cleanly explains (if it validates)

- Why leptons are cleaner (boundary = well-defined Shilov K-types) and quarks messier (interior bulk integrals + color + confinement).
- Why $m_s/m_d = \text{rank}^2 \cdot n_C$ uses n_C (the BULK dimension) — it’s a bulk/interior ratio.
- The top’s near-saturation as an interior-nearest-boundary effect, not literal boundary-reaching.

— Keeper K803, 2026-07-21. Diagnosis: capstone stalled because quarks were put at the BOUNDARY orbits like leptons. Casey: quark = CONFINED = interior bulk mode (rolling in curvature, not emitted); lepton = colorless boundary (Shilov). Reconnects to established bulk-vs-Shilov two-region (CLAUDE.md). Web: confinement = normalizable bulk modes; D_{IV}^5 boundedness = the IR cutoff. INVESTIGATE: redo the QUARK mass as a confined interior O-overlap $\langle f_n | O | f_n \rangle$ at $r^2 = k/(k+N_c)$ with color N_c — does it output $m_s/m_d=20$ tuning-free? Top = nearest-boundary confined quark (refines F603). THIRD reframe — gates UNCHANGED: (a) 20 tuning-free + (b) second quantity predicted; sub-check $k=C_2$ forced or fit; pivot-exit stands. See [[Keeper_K802_capstone_wellposed_but_UNVALIDATED_ms20_is_the_anchor_not_the_test_validation_criteria_2026-07-21]], CLAUDE.md bulk-vs-Shilov, Grace $r^2 = k/(k+N_c)$, Elie O-overlap.